Claude Agents SDK BEATS all Agent Framework! (Beginners Guide)

Mervin Praison
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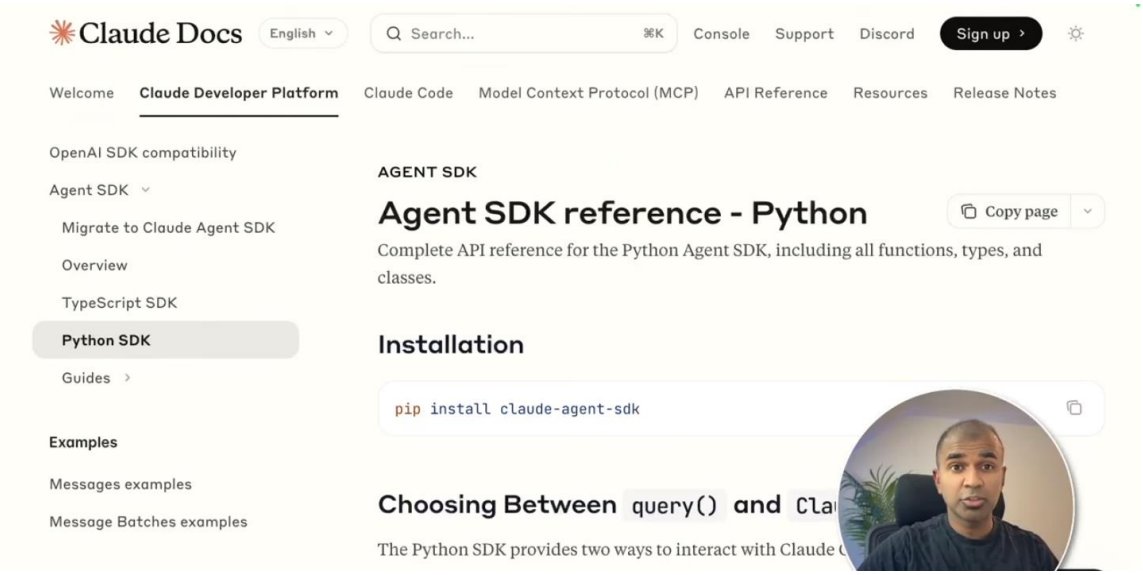
Claude Agents SDK: Build AI Agents with Claude Code Integration | Complete Tutorial

Learn how to harness the power of Claude Agents SDK to create sophisticated AI agents that can solve complex tasks autonomously. In this comprehensive tutorial, I'll show you how to integrate Claude Code—one of the most advanced CLI tools available—into your own Python applications.

https://mer.vin/2025/10/claude-agent-...
https://docs.claude.com/en/docs/claud...
https://docs.claude.com/en/api/agent-...

What You'll Learn:
✅ Install and set up Claude Agents SDK
✅ Create a basic AI agent from scratch
✅ Integrate internal tools (read/write permissions)
✅ Build custom tools and MCP servers
✅ Configure Claude Code options for advanced functionality
✅ Create a Python web server using AI agents

Timestamps:
0:00 - Introduction to Claude Agents SDK
0:44 - Installation & Setup
1:25 - Creating a Basic Agent
2:36 - Using Internal Tools (Read/Write)
4:01 - Creating Custom Tools with MCP Servers
5:35 - Advanced Claude Options Configuration
6:31 - Why Claude Agents SDK Stands Out



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Getting started
Overview
Quickstart
Common workflows

Build with Claude Code
Subagents
Output styles
Hooks
Headless mode
GitHub Actions
GitLab CI/CD
Model Context Protocol (MCP)

GETTING STARTED

Claude Code overview

Copy page

Learn about Claude Code, Anthropic's agentic coding tool that lives in your terminal and helps you turn ideas into code faster than ever before.

Get started in 30 seconds

Prerequisites:

- Node.js 18 or newer
- A Claude.ai (recommended) or Claude Console accoun

```
# Install Claude Code
npm install -g @anthropic-ai/claude-code

# Navigate to your project
```

What Claude Code does for you

- **Build features from descriptions**: Tell Claude what you want to build in plain English. will make a plan, write the code, and ensure it works.
- **Debug and fix issues**: Describe a bug or paste an error message. Claude Code will analy your codebase, identify the problem, and implement a fix.
- **Navigate any codebase**: Ask anything about your team's codebase, and get a thoughtfu answer back. Claude Code maintains awareness of your entire project structure, can fir up-to-date information from the web, and with MCP can pull f datasource like Google Drive, Figma, and Slack.
- **Automate tedious tasks**: Fix fiddly lint issues, resolve mer releas notes. Do all this in a single command from your developer atically in CI.

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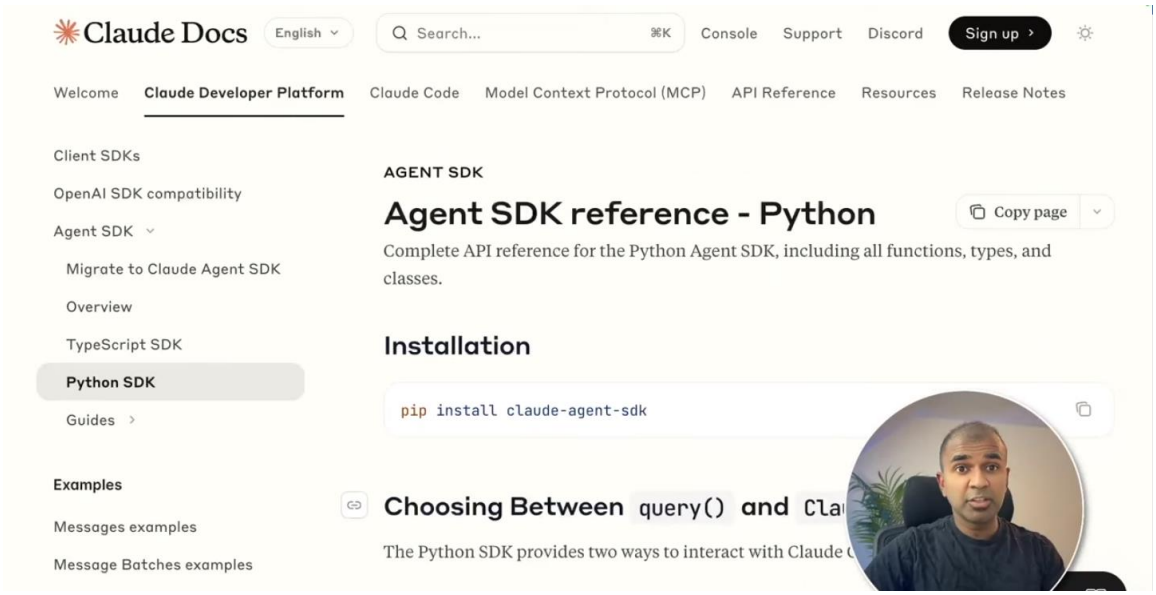
- **Automate tedious tasks**: Fix fiddly lint issues, resolve ase notes. Do all this in a single command from your devel ly in CI.



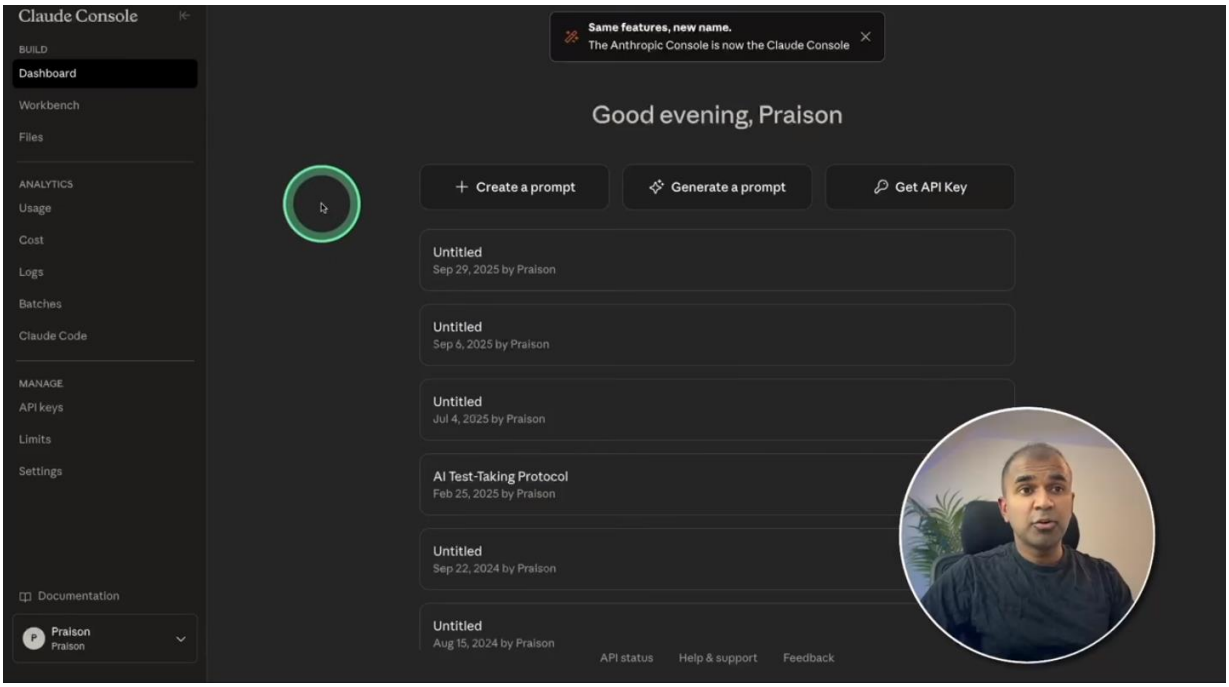

```
❯ npm i -g @anthropic-ai/claude-code

added 9 packages, and changed 3 packages in 1s

11 packages are looking for funding
  run `npm fund` for details
❯
```

```
❯ pip install claude-agent-sdk                                    Add to Chat ⌘L
Requirement already satisfied: claude-agent-sdk in /Users/praison/mini
conda3/envs/test/lib/python3.12/site-packages (0.1.0)
Requirement already satisfied: anyio>=4.0.0 in /Users/praison/minicond
a3/envs/test/lib/python3.12/site-packages (from claude-agent-sdk) (4.1
1.0)
Requirement already satisfied: mcp>=0.1.0 in /Users/praison/miniconda3
/envs/test/lib/python3.12/site-packages (from claude-agent-sdk) (1.16.
0)
```

```
❯ export ANTHROPIC_API_KEY=xxxxxxxxxxxxxxx
```



Get this key from platform.api.com

```
❯ python app.py
SystemMessage(
    subtype='init',
    data={
        'type': 'system',
        'subtype': 'init',
        'cwd': '/Users/praison/cc',
        'session_id': '8029ba33-052d-4950-8ec5-0c78c5a6387d',
        'tools': [
            'Task',
            'Bash',
            'Glob',
            'Grep',
            'ExitPlanMode',
            'Read',
            'Edit',
            'Write',
            'NotebookEdit',
            'WebFetch',
            'TodoWrite',
            'WebSearch',
            'BashOutput',
            'KillShell',
            'SlashCommand'
        ],
        'mcp_servers': [],
```

```
            'pr-comments',
            'release-notes',
            'todos',
            'review',
            'security-review'
        ],
        'apiKeySource': 'ANTHROPIC_API_KEY',
        'output_style': 'default',
        'agents': [
            'general-purpose',
            'statusline-setup',
            'output-style-setup'
        ],
        'uuid': '96a5dbd6-421d-43cd-acf2-95d76b1beced'
    }
)
AssistantMessage(
    content=[
        TextBlock(
            text="Hello! I'm doing well, thank you. How can I help you t
        )
    ],
    model='claude-sonnet-4-5-20250929',
    parent_tool_use_id=None
)
ResultMessage(
```

```
            'statusline-setup',
            'output-style-setup'
        ],
        'uuid': '96a5dbd6-421d-43cd-acf2-95d76b1beced'
    }
)
AssistantMessage(
    content=[
        TextBlock(
            text="Hello! I'm doing well, thank you. How can I help you today?"
        )
    ],
    model='claude-sonnet-4-5-20250929',
    parent_tool_use_id=None
)
ResultMessage(
    subtype='success',
    duration_ms=2748,
    duration_api_ms=2616,
    is_error=False,
    num_turns=1,
    session_id='8029ba33-052d-4950-8ec5-0c78c5a6387d',
    total_cost_usd=0.036115049999999996,
    usage={
        'input_tokens': 3,
        'cache_creation_input_tokens': 9125,
```

```
    model='claude-sonnet-4-5-20250929',
    parent_tool_use_id=None
)
ResultMessage(
    subtype='success',
    duration_ms=2748,
    duration_api_ms=2616,
    is_error=False,
    num_turns=1,
    session_id='8029ba33-052d-4950-8ec5-0c78c5a6387d',
    total_cost_usd=0.036115049999999996,
    usage={
        'input_tokens': 3,
        'cache_creation_input_tokens': 9125,
        'cache_read_input_tokens': 5291,
        'output_tokens': 20,
        'server_tool_use': {'web_search_requests': 0},
        'service_tier': 'standard',
        'cache_creation': {
            'ephemeral_1h_input_tokens': 0,
            'ephemeral_5m_input_tokens': 9125
        }
    },
    result="Hello! I'm doing well, thank you. How can I help you today?"
)
❯
```

2 Internal Tools

existing_tools.py

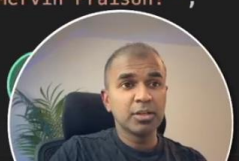
```python
import asyncio
from claude_agent_sdk import query, ClaudeAgentOptions

async def main():

    options = ClaudeAgentOptions(
        allowed_tools=["Read", "Write"],
        permission_mode="acceptEdits"
    )

    async for msg in query(
        prompt="Create a file called greeting.txt with 'Hello Mervin Praison!'",
        options=options
    ):
        print(msg)

asyncio.run(main())
```



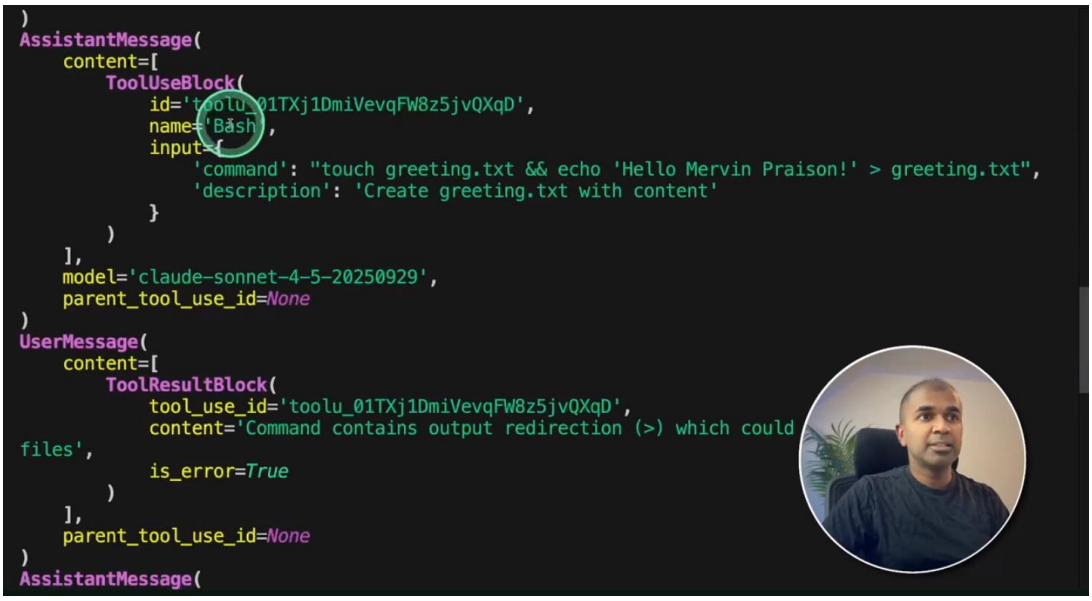
```
❯ python existing_tools.py
```

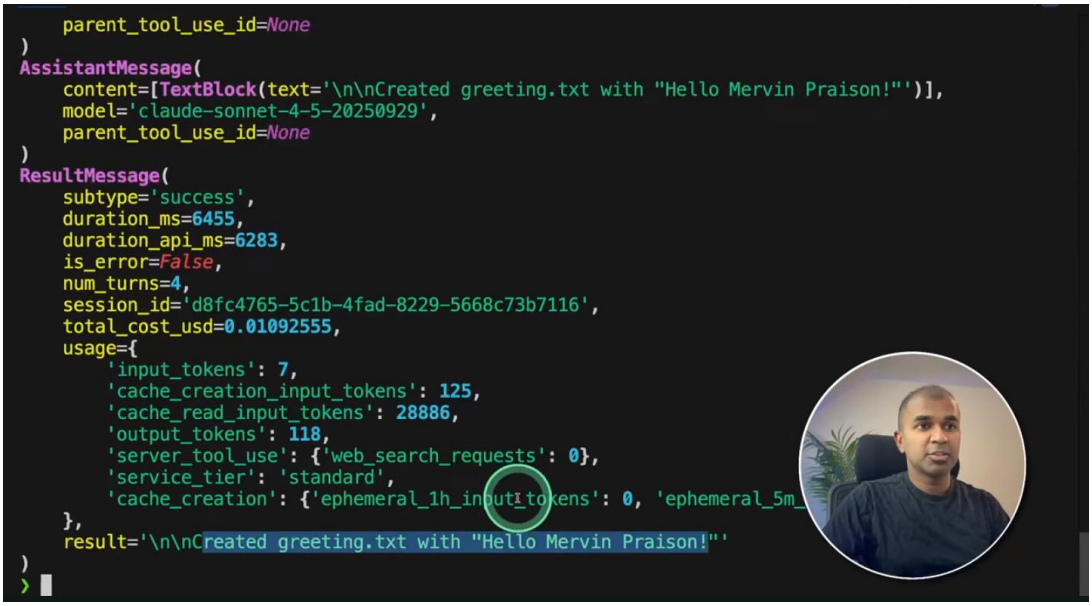
```
❯ python existing_tools.py
SystemMessage(
    subtype='init',
    data={
        'type': 'system',
        'subtype': 'init',
        'cwd': '/Users/praison/cc',
        'session_id': '78c59d5d-3128-4517-abe0-b5138d9087ce',
        'tools': [
            'Task',
            'Bash',
            'Glob',
            'Grep',
            'ExitPlanMode',
            'Read',
            'Edit',
            'Write',
            'NotebookEdit',
            'WebFetch',
            'TodoWrite',
            'WebSearch',
            'BashOutput',
            'KillShell',
            'SlashCommand'
        ],
        'mcp_servers': [],
        'model': 'claude-sonnet-4-5-20250929',
```

```
            'Glob',
            'Grep',
            'ExitPlanMode',
            'Read',
            'Edit',
            'Write',
            'NotebookEdit',
            'WebFetch',
            'TodoWrite',
            'WebSearch',
            'BashOutput',
            'KillShell',
            'SlashCommand'
        ],
        'mcp_servers': [],
        'model': 'claude-sonnet-4-5-20250929',
        'permissionMode': 'acceptEdits',
        'slash_commands': [
            'compact',
            'context',
            'cost',
            'init',
            'output-style:new',
            'pr-comments',
            'release-notes',
            'todos',
            'review',
```

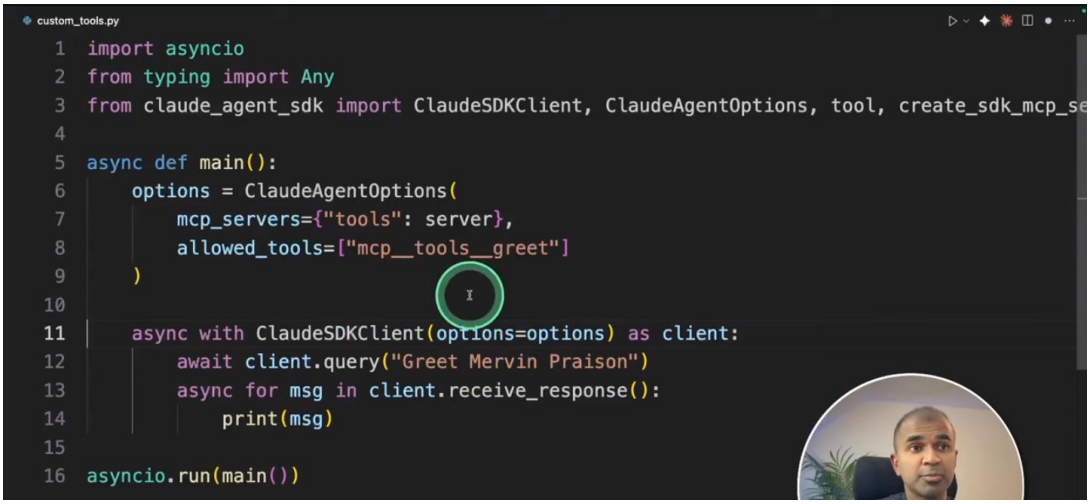
```
        ],
        'mcp_servers': [],
        'model': 'claude-sonnet-4-5-20250929',
        'permissionMode': 'acceptEdits',
        'slash_commands': [
            'compact',
            'context',
            'cost',
            'init',
            'output-style:new',
            'pr-comments',
            'release-notes',
            'todos',
            'review',
            'security-review'
        ],
        'apiKeySource': 'ANTHROPIC_API_KEY',
        'output_style': 'default',
        'agents': ['general-purpose', 'statusline-setup', 'output-style-se
        'uuid': 'd86a08f7-604a-418a-a261-546d482caf7d'
    }
)
AssistantMessage(
    content=[
        ToolUseBlock(
            id='toolu_01NafKxyxP6ZphjFJKtB7xaw',
            name='Write',
```

```
            'todos',
            'review',
            'security-review'
        ],
        'apiKeySource': 'ANTHROPIC_API_KEY',
        'output_style': 'default',
        'agents': ['general-purpose', 'statusline-setup', 'output-style-setup'],
        'uuid': 'd86a08f7-604a-418a-a261-546d482caf7d'
    }
)
AssistantMessage(
    content=[
        ToolUseBlock(
            id='toolu_01NafKxyxP6ZphjFJKtB7xaw',
            name='Write',
            input={
                'file_path': '/Users/praison/cc/greeting.txt',
                'content': 'Hello Mervin Praison!'
            }
        )
    ],
    model='claude-sonnet-4-5-20250929',
    parent_tool_use_id=None
)
UserMessage(
    content=[
        ToolResultBlock(
```

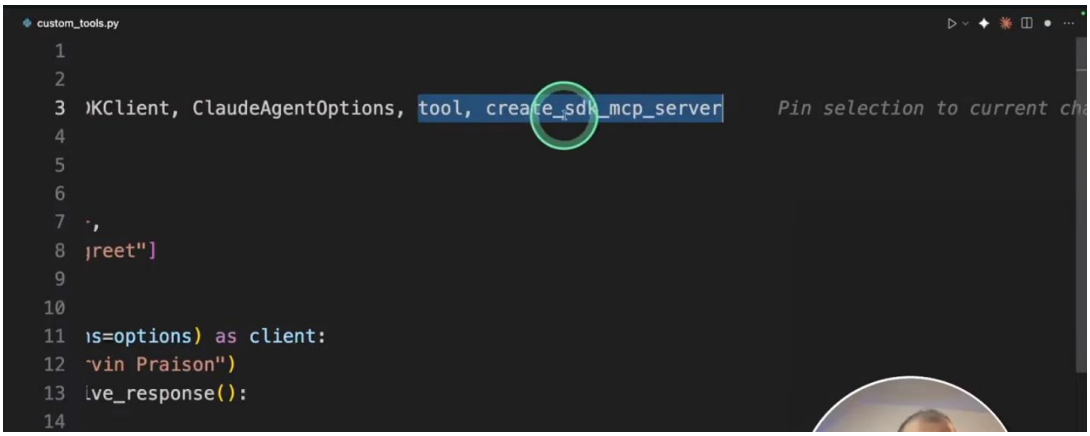


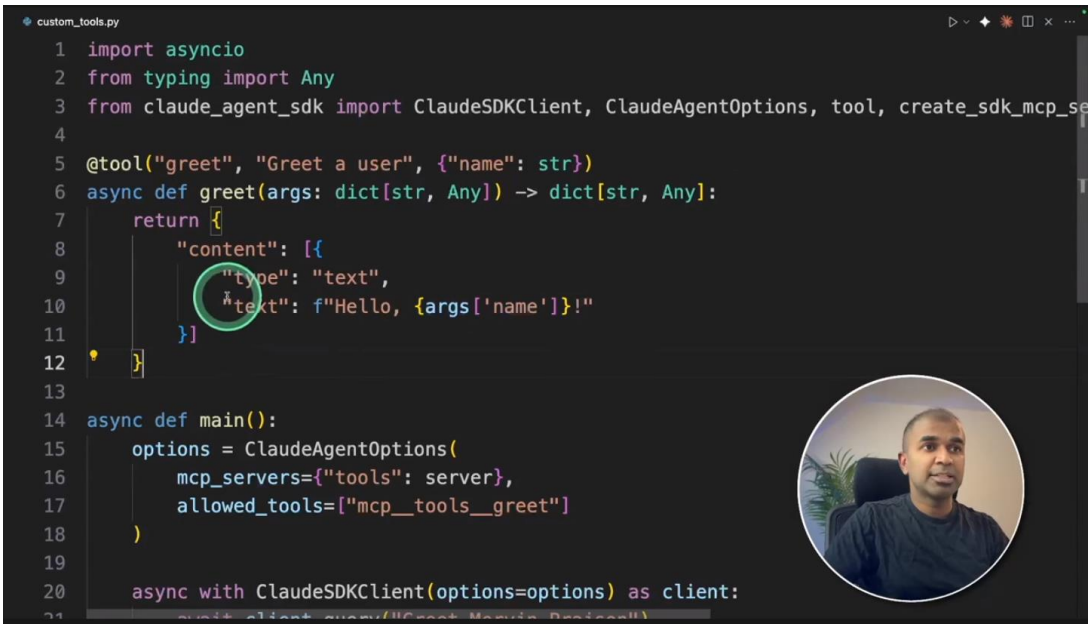


The file was written successfully using the internally available ‘Write’ tool



To add a custom tool, you need to set it up as an MCP server





We then create a new tool called **greet** as above that we can now integrate with any API

custom_tools.py

```python
async def greet(args: dict[str, Any]) -> dict[str, Any]:
        "content": [{
            "type": "text",
            "text": f"Hello, {args['name']}!"
        }]
    }

server = create_sdk_mcp_server(
    name="my-tools",
    version="1.0.0",
    tools=[greet]
)

async def main():
    options = ClaudeAgentOptions(
        mcp_servers={"tools": server},     Pin selection to cur                    d+Opti
        allowed_tools=["mcp__tools__greet"]
    )

    async with ClaudeSDKClient(options=options) as client:
```

custom_tools.py

```python
server = create_sdk_mcp_server(
    version="1.0.0",
    tools=[greet]
)

async def main():
    options = ClaudeAgentOptions(
        mcp_servers={"tools": server},
        allowed_tools=["mcp__tools__greet"]
    )

    async with ClaudeSDKClient(options=options) as client:
        await client.query("Greet Mervin Praison")
        async for msg in client.receive_response():
            print(msg)

asyncio.run(main())
```

TERMINAL

```
❯ python custom_tools.py
```

TERMINAL

```
    subtype='init',
    data={
        'type': 'system',
        'subtype': 'init',
        'cwd': '/Users/praison/cc',
        'session_id': '33e726f0-fd20-4d4f-ab97-a508a0ea40f0',
        'tools': [
            'Task',
            'Bash',
            'Glob',
            'Grep',
            'ExitPlanMode',
            'Read',
            'Edit',
            'Write',
            'NotebookEdit',
            'WebFetch',
            'TodoWrite',
            'WebSearch',
            'BashOutput',
            'KillShell',
            'SlashCommand',
            'mcp__tools__greet'
        ],
        'mcp_servers': [{'name': 'tools', 'status': 'connected'}],
        'model': 'claude-sonnet-4-5-20250929',
        'permissionMode': 'default',
        'slash_commands': [
            'compact',
            'context',
            'cost',
            'init',
```

TERMINAL

```
            'context',
            'cost',
            'init',
            'output-style:new',
            'pr-comments',
            'release-notes',
            'todos',
            'review',
            'security-review'
        ],
        'apiKeySource': 'ANTHROPIC_API_KEY',
        'output_style': 'default',
        'agents': ['general-purpose', 'statusline-setup', 'output-style-setup'],
        'uuid': '819491e0-9d23-4c29-8104-05aba1ff494d'
    }
)
AssistantMessage(
    content=[
        ToolUseBlock(
            id='toolu_016NfuVUTuKFNKLeTSmuPQwA',
            name='mcp__tools__greet',
            input={'name': 'Mervin Praison'}
        )
    ],
    model='claude-sonnet-4-5-20250929',
    parent_tool_use_id=None
)
UserMessage(
    content=[
        ToolResultBlock(
            tool_use_id='toolu_016NfuVUTuKFNKLeTSmuPQwA',
            content=[{'type': 'text', 'text': 'Hello, Mervin Praison!'}],
```

TERMINAL

```
        'output_style': 'default',
        'agents': ['general-purpose', 'statusline-setup', 'output-style-setup'],
        'uuid': '819491e0-9d23-4c29-8104-05aba1ff494d'
    }
)
AssistantMessage(
    content=[
        ToolUseBlock(
            id='toolu_016NfuVUTuKFNKLeTSmuPQwA',
            name='mcp__tools__greet',
            input={'name': 'Mervin Praison'}
        )
    ],
    model='claude-sonnet-4-5-20250929',
    parent_tool_use_id=None
)
UserMessage(
    content=[
        ToolResultBlock(
            tool_use_id='toolu_016NfuVUTuKFNKLeTSmuPQwA',
            content=[{'type': 'text', 'text': 'Hello, Mervin Praison!'}],
            is_error=None
        )
    ],
    parent_tool_use_id=None
)
AssistantMessage(
    content=[TextBlock(text='\n\nHello, Mervin Praison!')],
    model='claude-sonnet-4-5-20250929',
    parent_tool_use_id=None
)
ResultMessage(
```

```
TERMINAL
            tool_use_id='toolu_016NfuVUTuKFNKLeTSmuPQwA',
            content=[{'type': 'text', 'text': 'Hello, Mervin Praison!'}],
            is_error=None
        )
    ],
    parent_tool_use_id=None
)
AssistantMessage(
    content=[TextBlock(text='\n\nHello, Mervin Praison!')],
    model='claude-sonnet-4-5-20250929',
    parent_tool_use_id=None
)
ResultMessage(
    subtype='success',
    duration_ms=6318,
    duration_api_ms=7199,
    is_error=False,
    num_turns=3,
    session_id='33e726f0-fd20-4d4f-ab97-a508a0ea40f0',
    total_cost_usd=0.042039950000000006,
    usage={
        'input_tokens': 6,
        'cache_creation_input_tokens': 9285,
        'cache_read_input_tokens': 19786,
        'output_tokens': 75,
        'server_tool_use': {'web_search_requests': 0},
        'service_tier': 'standard',
        'cache_creation': {'ephemeral_1h_input_tokens': 0, 'ephemeral_5m_input_tokens':
    },
    result='\n\nHello, Mervin Praison!'
)
❯
```

```
custom_tools.py
1  import asyncio
2  from typing import Any
3  from claude_agent_sdk import ClaudeSDKClient, ClaudeAgentOptions, tool, create_sdk_mcp_server
4  from rich import print
5
6  @tool("greet", "Greet a user", {"name": str})
7  async def greet(args: dict[str, Any]) -> dict[str, Any]:
8      return {
9          "content": [{
10             "type": "text",
11             "text": f"Hello, {args['name']}!"
12         }]
13     }
14
15 server = create_sdk_mcp_server(
16     name="my-tools",
17     version="1.0.0",
18     tools=[greet]
19 )
20
21 async def main():
22     options = ClaudeAgentOptions(
23         mcp_servers={"tools": server},
24         allowed_tools=["mcp__tools__greet"]
25     )
```

Claude Options

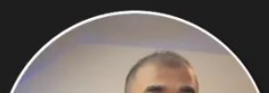
create_server.py

```python
import asyncio
from claude_agent_sdk import query, ClaudeAgentOptions

async def main():
    options = ClaudeAgentOptions(
        system_prompt="You are an expert Python developer",
        permission_mode='acceptEdits',
        cwd="/Users/praison/cc"
    )

    async for message in query(
        prompt="Create a Python web server",
        options=options
    ):
        print(message)

asyncio.run(main())
```



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- OpenAI SDK compatibility
- Agent SDK
  - Migrate to Claude Agent SDK
  - Overview
  - TypeScript SDK
  - Python SDK
  - Guides
- Examples
  - Messages examples
  - Message Batches examples

AGENT SDK

Agent SDK reference - Python

Copy page

Complete API reference for the Python Agent SDK, including all functions, types, and classes.

Installation

```
pip install claude-agent-sdk
```

Choosing Between `query()` and `Cla`

The Python SDK provides two ways to interact with Claude

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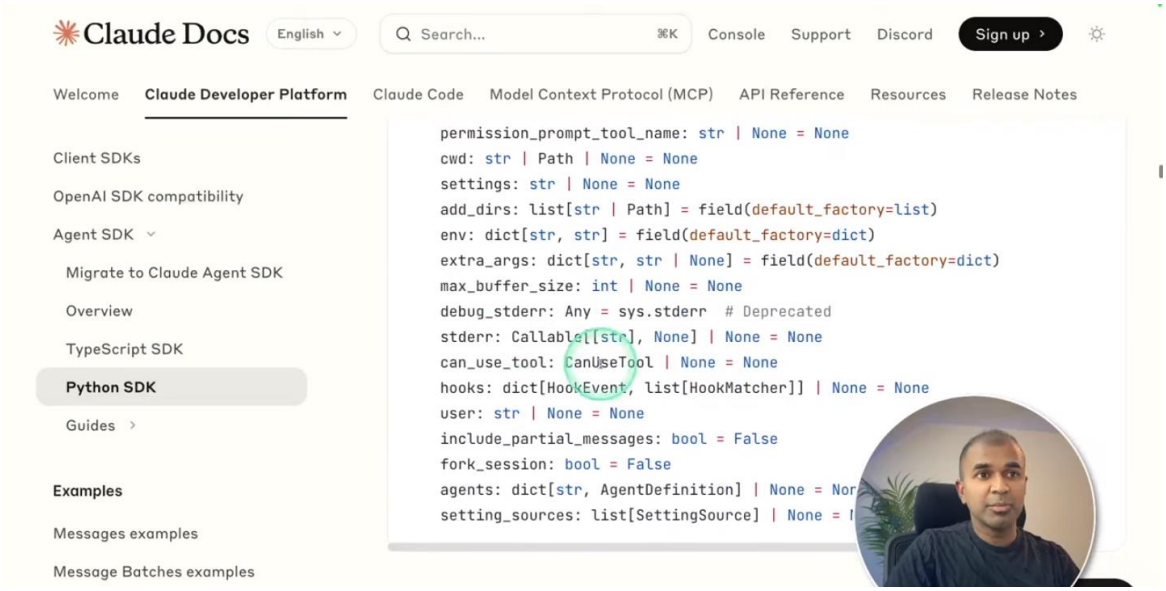
  - Message Batches examples
- 3rd-party APIs

Any]]] execution

ClaudeAgentOptions

Configuration dataclass for Claude Code queries.

```python
@dataclass
class ClaudeAgentOptions:
    allowed_tools: list[str] = field(default_factory=list)
    system_prompt: str | SystemPromptPreset | None = None
    mcp_servers: dict[str, McpServerConfig] | str | Pa
    permission_mode: PermissionMode | None = None
    continue_conversation: bool = False
    resume: str | None = None
    max_turns: int | None = None
    disallowed_tools: list[str] = field(default_fac
    model: str | None = None
    permission_prompt_tool_name: str | None = None
    cwd: str | Path | None = None
```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Discord
Sign up
Welcome
Claude Developer Platform
Claude Code
Model Context Protocol (MCP)
API Reference
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OpenAI SDK compatibility
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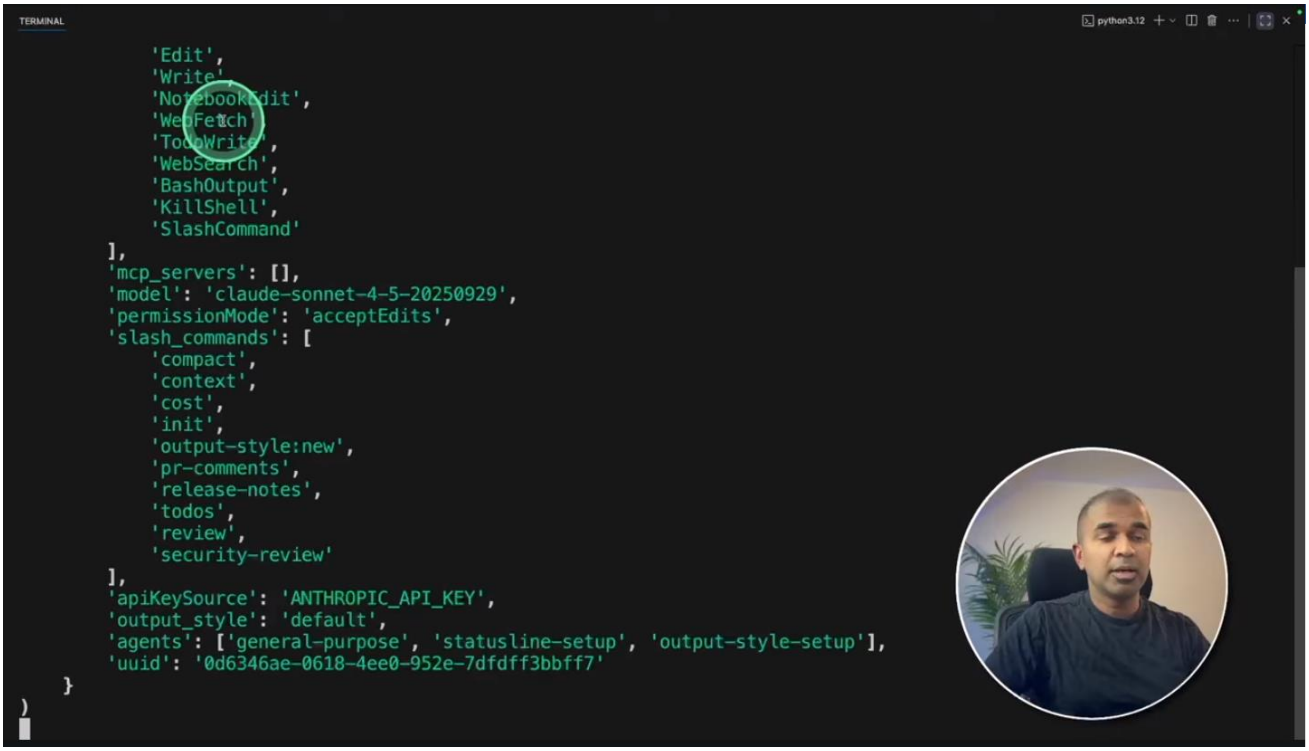
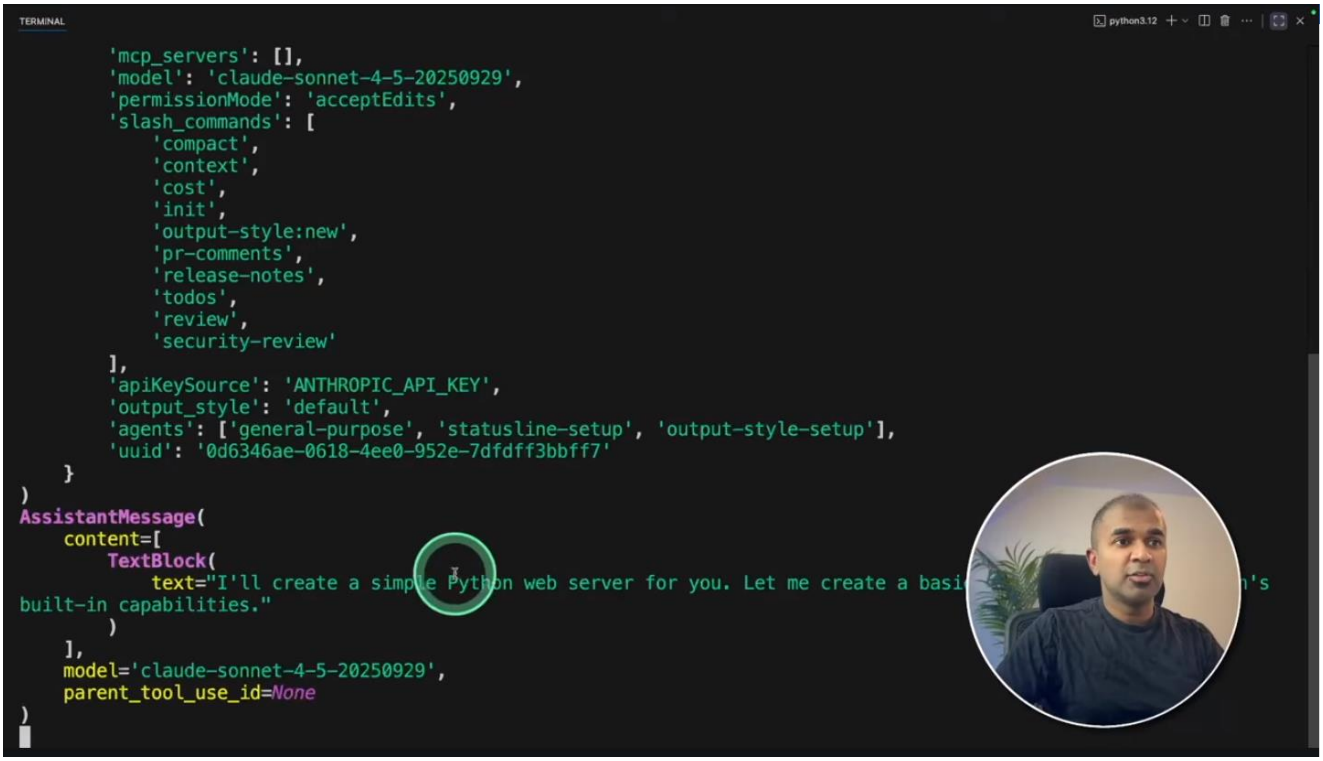
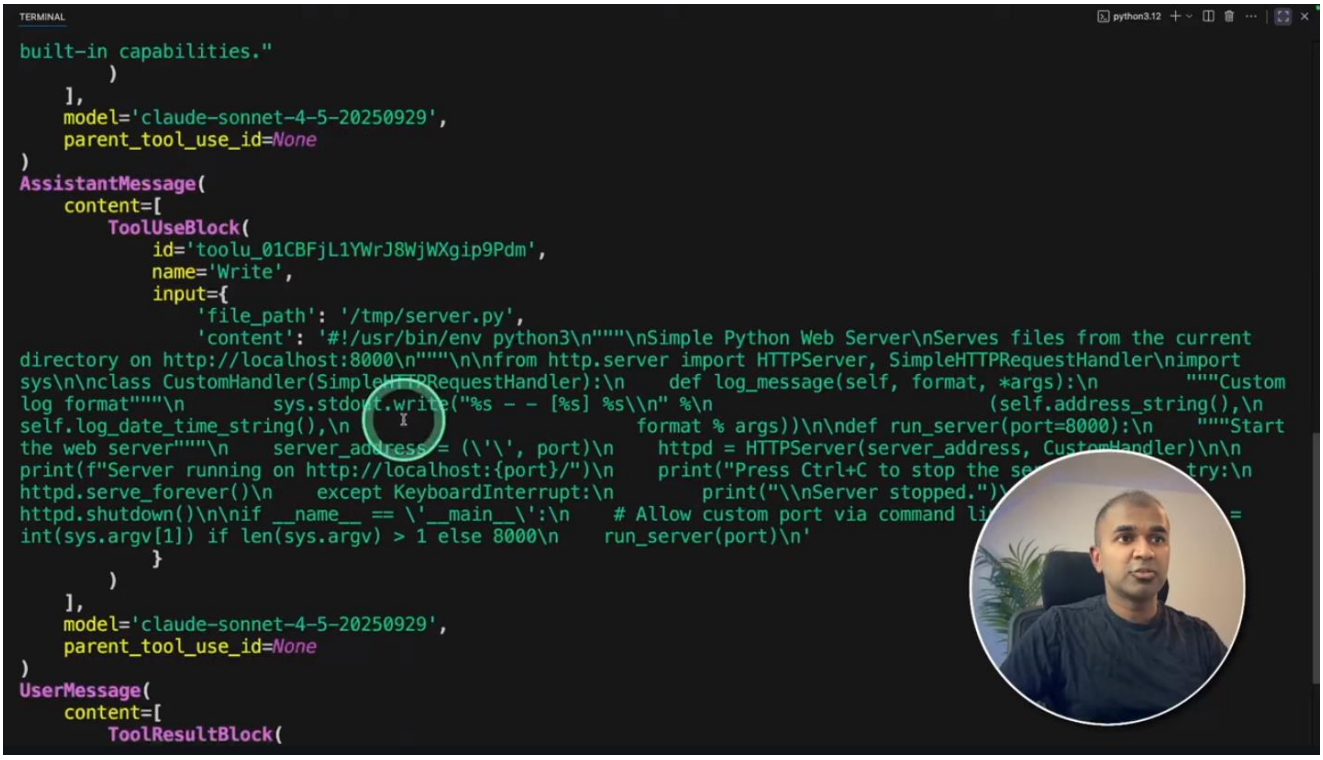
Examples
Messages examples
Message Batches examples
permission_prompt_tool_name: str | None = None
cwd: str | Path | None = None
settings: str | None = None
add_dirs: list[str | Path] = field(default_factory=list)
env: dict[str, str] = field(default_factory=dict)
extra_args: dict[str, str | None] = field(default_factory=dict)
max_buffer_size: int | None = None
debug_stderr: Any = sys.stderr # Deprecated
stderr: Callable[[str], None] | None = None
can_use_tool: CanUseTool | None = None
hooks: dict[HookEvent, list[HookMatcher]] | None = None
user: str | None = None
include_partial_messages: bool = False
fork_session: bool = False
agents: dict[str, AgentDefinition] | None = Nor
setting_sources: list[SettingSource] | None = I


❯ python create_server.py


TERMINAL
python3.12
'Edit',
'Write',
'NotebookEdit',
'WebFetch',
'TodoWrite',
'WebSearch',
'BashOutput',
'KillShell',
'SlashCommand'
],
'mcp_servers': [],
'model': 'claude-sonnet-4-5-20250929',
'permissionMode': 'acceptEdits',
'slash_commands': [
'compact',
'context',
'cost',
'init',
'output-style:new',
'pr-comments',
'release-notes',
'todos',
'review',
'security-review'
],
'apiKeySource': 'ANTHROPIC_API_KEY',
'output_style': 'default',
'agents': ['general-purpose', 'statusline-setup', 'output-style-setup'],
'uuid': '0d6346ae-0618-4ee0-952e-7dfdff3bbff7'
}
)


TERMINAL
python3.12
'mcp_servers': [],
'model': 'claude-sonnet-4-5-20250929',
'permissionMode': 'acceptEdits',
'slash_commands': [
'compact',
'context',
'cost',
'init',
'output-style:new',
'pr-comments',
'release-notes',
'todos',
'review',
'security-review'
],
'apiKeySource': 'ANTHROPIC_API_KEY',
'output_style': 'default',
'agents': ['general-purpose', 'statusline-setup', 'output-style-setup'],
'uuid': '0d6346ae-0618-4ee0-952e-7dfdff3bbff7'
}
)
AssistantMessage(
content=[
TextBlock(
text="I'll create a simple Python web server for you. Let me create a basi
's
built-in capabilities."
)
],
model='claude-sonnet-4-5-20250929',
parent_tool_use_id=None
)


TERMINAL
python3.12
built-in capabilities."
)
],
model='claude-sonnet-4-5-20250929',
parent_tool_use_id=None
)
AssistantMessage(
content=[
ToolUseBlock(
id='toolu_01CBFjL1YWrJ8WjWXgip9Pdm',
name='Write',
input={
'file_path': '/tmp/server.py',
'content': '#!/usr/bin/env python3\n"""\nSimple Python Web Server\nServes files from the current
directory on http://localhost:8000\n"""\n\nfrom http.server import HTTPServer, SimpleHTTPRequestHandler\nimport
sys\n\nclass CustomHandler(SimpleHTTPRequestHandler):\n def log_message(self, format, *args):\n """Custom
log format"""\n sys.stdout.write("%s - - [%s] %s\\n" %\n (self.address_string(),\n
self.log_date_time_string(),\n format % args))\n\ndef run_server(port=8000):\n """Start
the web server"""\n server_address = (\'\', port)\n httpd = HTTPServer(server_address, CustomHandler)\n\n
print(f"Server running on http://localhost:{port}/")\n print("Press Ctrl+C to stop the se try:\n
httpd.serve_forever()\n except KeyboardInterrupt:\n print("\\nServer stopped.")\
httpd.shutdown()\n\nif __name__ == \'__main__\':\n # Allow custom port via command li =
int(sys.argv[1]) if len(sys.argv) > 1 else 8000\n run_server(port)\n'
}
)
],
model='claude-sonnet-4-5-20250929',
parent_tool_use_id=None
)
UserMessage(
content=[
ToolResultBlock(

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```
AssistantMessage(
    content=[
        ToolUseBlock(
            id='toolu_01TJrb379a7uYEQMPmeHJbg5',
            name='Write',
            input={
                'file_path': 'server.py',
                'content': '#!/usr/bin/env python3\n"""\nSimple HTTP Web Server\nRun with: python
server.py\n"""\n\nfrom http.server import HTTPServer, SimpleHTTPRequestHandler\nimport sys\n\ndef
run_server(port=8000):\n    """Start the web server on the specified port."""\n    server_address = (\'\', port)\n
httpd = HTTPServer(server_address, SimpleHTTPRequestHandler)\n\n    print(f"Server running on
http://localhost:{port}/")\n    print("Press Ctrl+C to stop the server")\n\n    try:\n        httpd.serve_forever()\n
except KeyboardInterrupt:\n        print("\\nServer stopped.")\n        sys.exit(0)\n\nif __name__ == "__main__":\n
port = 8000\n    if len(sys.argv) > 1:\n        try:\n            port = int(sys.argv[1])\n        except
ValueError:\n            print("Invalid port number. Using default port 8000.")\n\n    run_server(port)\n'
            }
        )
    ],
    model='claude-sonnet-4-5-20250929',
    parent_tool_use_id=None
)
UserMessage(
    content=[
        ToolResultBlock(
            tool_use_id='toolu_01TJrb379a7uYEQMPmeHJbg5',
            content='File created successfully at: server.py',
            is_error=None
        )
    ],
    parent_tool_use_id=None
)
```

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```
default.\n\n**To use a custom port:**\n```bash\npython server.py 3000\n```\n\nThe server will serve files from the
current directory and can be stopped with `Ctrl+C`. It uses Python's built-in `http.server` module, so no additional
dependencies are required."
        )
    ],
    model='claude-sonnet-4-5-20250929',
    parent_tool_use_id=None
)
ResultMessage(
    subtype='success',
    duration_ms=12376,
    duration_api_ms=12193,
    is_error=False,
    num_turns=4,
    session_id='0a157186-87d6-404c-8c75-53f74dbb92e8',
    total_cost_usd=0.018286649999999998,
    usage={
        'input_tokens': 9,
        'cache_creation_input_tokens': 1277,
        'cache_read_input_tokens': 22453,
        'output_tokens': 449,
        'server_tool_use': {'web_search_requests': 0},
        'service_tier': 'standard',
        'cache_creation': {'ephemeral_1h_input_tokens': 0, 'ephemeral_5m_input_tokens'
    },
    result="\n\nI've created a Python web server file called `server.py` in your curre                    the
server:**\n```bash\npython server.py\n```\n\nThis will start the server on `http://loca
default.\n\n**To use a custom port:**\n```bash\npython server.py 3000\n```\n\nThe serve                    the
current directory and can be stopped with `Ctrl+C`. It uses Python's built-in `http.serve                    itional
dependencies are required."
)
```

server.py

```python
#! /usr/bin/env python3
"""
Simple HTTP Web Server
Run with: python server.py
"""

from http.server import HTTPServer, SimpleHTTPRequestHandler
import sys


def run_server(port=8000):
    """Start the web server on the specified port."""
    server_address = ('', port)
    httpd = HTTPServer(server_address, SimpleHTTPRequestHandler)

    print(f"Server running on http://localhost:{port}/")
    print("Press Ctrl+C to stop the server")

    try:
        httpd.serve_forever()
    except KeyboardInterrupt:
        print("\nServer stopped.")
        sys.exit(0)

```

server.py

```python
def run_server(port=8000):
    print("Press Ctrl+C to stop the server")

    try:
        httpd.serve_forever()
    except KeyboardInterrupt:
        print("\nServer stopped.")
        sys.exit(0)


if __name__ == "__main__":
    port = 8000
    if len(sys.argv) > 1:
        try:
            port = int(sys.argv[1])
        except ValueError:
            print("Invalid port number. Using default port 8000.")

    run_server(port)

```

Above is the generated python code that was created for us.